AI 101 for Mechanical Engineers · Module 8

Imbalance and deployment consequences

Lesson 8.3

What you will learn

> Explain why class imbalance can make accuracy misleading, evaluate a threshold on separate data, and identify the evidence needed before acting on alerts.

Lesson overview

Module 8 · Lesson 3

Imbalance and deployment consequences

AI 101 for Mechanical Engineers

1 Accuracy can hide missed faults

Imbalanced example

99 healthy	1 fault
operating cases	a rare fault case

Baseline: always predict healthy

Always healthy for every case → **Accuracy** 99% → **Fault recall** 0%

The rule misses the only fault.

Possible responses: collect representative fault examples, use class weights or resample the training data. These are not guaranteed fixes, and resampling belongs inside training folds only.

2 Select, then test

Validation
Choose the threshold using false-alarm and missed-fault consequences.

Separate fixed test
Evaluate the selected threshold without retuning it there.

Report more than accuracy

1. Positive class and class counts
2. Threshold and how it was selected
3. TP, FP, FN and TN
4. Precision, recall and operating conditions

Lab 3 uses synthetic scores, not a calibrated machinery detector.

3 Define what happens after an alert

Alert
The model signals a possible fault.

→ **Evidence review**
Check data and inspect as needed.

→ **Authorized decision**
Continue, schedule maintenance or stop equipment.

State who receives alerts, who checks evidence, who may stop equipment and who records the decision.

False alarms
can use inspection resources and cause downtime.

Missed faults
can leave damage or hazards undetected.

The costs depend on the machinery and its intended use.

Key takeaway

A successful teaching exercise does not approve deployment.
Real use needs operating-domain evidence, procedures and authorization.

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Rare faults can make accuracy misleading

- Imbalance: some classes much more frequent than others [44]
- Faults may be rare compared with healthy operation
- Always predicting healthy (99 healthy, 1 fault): 99% accuracy
- It detects no actual faults: fault recall is 0%

> Compare with this majority-class baseline and inspect positive-class performance.

Imbalance responses are choices, not guaranteed fixes

- Options: more representative fault examples, class weighting, resampling
- Resample only within training folds
- Keep validation and test representative of intended use
- Check that synthetic examples remain physically plausible

> Threshold choice is a separate decision reflecting false-alarm and missed-fault consequences.

Select the threshold, then test separately

1. Record TP, FP, FN and TN for the decisions
2. Select the threshold on validation examples
3. Consider false alarms and missed faults
4. Evaluate on a separate fixed test population
5. Report class counts and selected threshold with metrics

The two errors have different consequences

False alarm

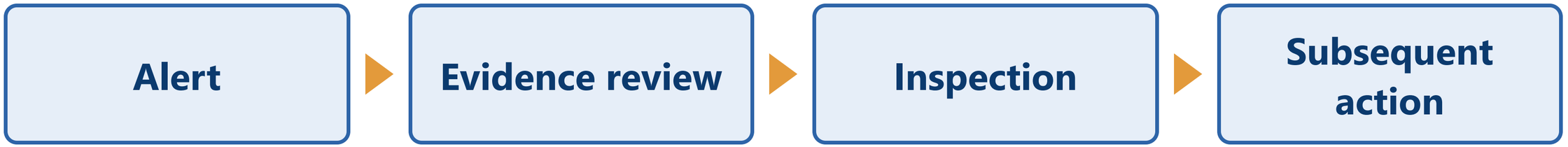
- May consume inspection resources
- Can cause avoidable downtime

Missed fault

- May leave damage undetected
- Can leave hazards undetected

> Costs depend on the machinery and use; the classroom example assigns no universal costs.

An alert starts an engineering procedure



- State who receives the alert
- State when an inspection is required
- State the evidence needed for a maintenance decision

Synthetic exercises do not authorize deployment

- Lab 3 uses synthetic scores and labels; not a trained or calibrated detector
- Synthetic demonstrations do not validate equipment safety
- A good exercise threshold needs further operating-domain evidence

Report more than aggregate accuracy

- Positive class and evaluated class counts
- Selected threshold and how it was chosen
- Missed faults and false alarms, plus precision and recall
- Evaluated population, alert recipients and evidence before action

Define accountability before acting on an alert

- Define who receives alerts and who checks evidence [9]
- Define who may stop equipment and records decisions [9]
- A score alone should not silently trigger consequential action
- Review thresholds with people responsible for the equipment

> Unless that use has been separately justified, tested and authorized.

CHECK YOUR UNDERSTANDING

Why can a detector with 99% accuracy be useless for rare faults?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why can a detector with 99% accuracy be useless for rare faults?

> Predicting all 100 examples as healthy labels the 99 healthy cases correctly but misses the only fault. Its fault recall is 0%.

CHECK YOUR UNDERSTANDING

Does a successful synthetic threshold exercise approve machinery deployment?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a successful synthetic threshold exercise approve machinery deployment?

> No. The exercise demonstrates evaluation. Real use needs appropriate evidence, inspection procedures and authorization.

Key takeaways

- Compare rare-fault detection against the majority-class baseline
- Select thresholds on validation; evaluate a separate fixed test
- Report counts and error consequences alongside metrics
- Define the alert-to-inspection procedure and evidence for action

References

[9] National Institute of Standards and Technology, “Artificial Intelligence Risk Management Framework (AI RMF 1.0),” NIST AI 100-1, Gaithersburg, MD, USA, 2023. doi: 10.6028/NIST.AI.100-1

[44] H. He and E. A. Garcia, “Learning from imbalanced data,” IEEE Trans. Knowl. Data Eng., vol. 21, no. 9, pp. 1263–1284, 2009. doi: 10.1109/TKDE.2008.239

Numbers match the course reference list. The written lesson is the authoritative text.